

AI-Driven Radiology Report Generation, Clinical NLP, and Healthcare Interoperability: A Comprehensive Survey

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Abstract—The modern radiology and the electronic health record (EHR) landscape are undergoing significant shifts with the inclusion of Artificial Intelligence(AI), Natural Language Processing (NLP) and informatics. The survey consolidates 27 reviewed publications from a range of automated radiology report generation works, multimodal vision-language models, clinical NLP, health care interoperability, semantic standardization and ethical considerations of AI deployment. The literature reviewed is summarized into four broad thematic groups: AI for Radiology Reporting and Multimodal Learning, AI for NLP in Clinical Practice and Patient Focused Applications, Healthcare Interoperability and Semantic Standardization, and AI Ethics and Reporting Transparency. Recent progress shows that Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs) in language-related tasks for report generation and clinical information extraction. Healthcare interoperability has greatly benefited from FHIR-based interoperability frameworks and ontology-based semantic systems. However, the critical issues, such as hallucination, cross-institutional generalization, semantic inconsistency, and lack of standards of clinical evaluation, still limit the widespread clinical use. This survey reveals the current gaps in research and proposes avenues for further development of clinically reliable, explainable, interoperable, and ethical AI systems in radiology.

Index Terms—Radiology Report Generation, Large Language Models, FHIR Interoperability, Vision-Language Models, Clinical NLP, Healthcare Ontologies, EHR.

I. INTRODUCTION

AI has proved to be one of the most impactful technologies in medical imaging and healthcare informatics. One pivotal area where AI automation is proving to be a game-changer is in radiology, a field that heavily depends on interpreting images and writing reports. With the increasing complexity in imaging exams and complex reporting, the development of automated report generation systems that can assist radiologists in generating reports and reduce the burden on their cognitive load has become a very important motivation. Recent progress in Transformer-based architectures and LLMs in the field of medical imaging has greatly enhanced the ability of artificial intelligence systems to interpret medical images and text. CNN-based and LSTM-based encoder-decoder systems

have gradually been replaced by more complex multimodal models that include large-scale language understanding and retrieval-augmented generation, which have been shown to perform surprisingly well in generating radiology reports and in supporting clinical decision-making processes, but have suffered from issues of hallucination, explainability, and diagnostic reliability [1]–[4]. In addition to report generation, the interoperability problem is a growing challenge in the healthcare industry, due to the presence of heterogeneous EHR systems, medical terminologies, and clinical infrastructures. At the same time, technologies like HL7 FHIR, openEHR, SNOMED CT and RadLex have become key to achieve semantic consistency, and new technology frameworks and solutions based on the semantic web are being developed to facilitate the integration of different biomedical knowledge sources and enable precision medicine applications [10]–[13]. The field of Clinical Natural Language Processing has also advanced significantly, ranging from information extraction and disease phenotyping to producing patient-friendly summaries and maintaining patient care systems using LLM-powered solutions [12]–[15]. These technologies help make health care more accessible and engaging for patients, but they also need to be accurate and under the supervision of clinicians if they are to be deployed safely. The ethical considerations of AI in the medical field have been growing and gaining more attention. Research focused on AI-generated patient summaries found that there are certain compromises between readability and clinical accuracy [16]. At the same time, the STARD-AI framework establishes standard reporting guidelines that will enhance transparency, fairness and reproducibility in AI diagnostic research [17].

The main results of this survey are briefly summarized below:

A single taxonomy for 27 peer-reviewed studies in four important research areas. A unified treatment of the aspect of report generation with AI, multimodal learning, frameworks for interoperability, and ethical practices in AI. Common limitations that are identified: hallucination, semantic incon-

sistencies, and generalization challenges. Discussion of future research directions towards clinically safe and interoperable multimodal healthcare systems.

II. LITERATURE SURVEY

The 27 reviewed studies are summarized under four themes.

Category I: AI Based Radiology Report Generation and Multimodal Models

These studies are related to automated radiology report generation, biomedical text simplification, multimodal vision-language architectures, and multimodal large language models. The shift from traditional encoder-decoder models to Transformer-based multimodal systems and retrieval-augmented architectures are explored in Papers [?], [1]–[3], [5], [18]–[23].

This course is for those who are interested in developing patient-centric applications and clinical NLP skills.

These papers focus on information extraction, rare disease detection, cross-institutional model transfer, patient-friendly note creation, and incorporating LLMs in healthcare administration and clinical decision-making.

IOM Standards for Semantic Standardization and Healthcare Interoperability

Research on FHIR-based interoperability, semantic mapping, ontology integration, privacy-preserving architectures, and biomedical knowledge representation focuses on FHIR-based interoperability, semantic mapping, ontology integration, privacy-preserving architectures and biomedical knowledge representation (see [6]–[11], [25]–[27]).

AI Ethics, Transparency, and Reporting Standards (Category IV): Ethical considerations, transparency, and ethical reporting.

The study of Papers [16], [17] explores ethical issues related to AI-generated clinical content and introduces consistent reporting practices to promote fairness, transparency, and responsible use of AI.

The taxonomy resulting from this work offers a comprehensive depiction of the existing ecosystem of multimodal AI in radiology and reveals the interconnected nature of the different aspects of report generation, clinical NLP, interoperability, and ethics governance.

III. AI-BASED RADIOLOGY REPORT GENERATION AND MULTIMODAL MODELS

AI-driven radiology report generation has moved beyond traditional image-captioning models to more complex multimodal AI systems that incorporate medical images, clinical text, external knowledge bases, and large language models. The reviewed studies have made substantial advances in linguistic quality, in support for diagnosis and in efficiency of workflow; however, issues of hallucination, clinical reliability and assessment criteria are not solved.

A. LLM-Based Pathology Labeling and Biomedical Text Simplification

Abdullah et al. [18] developed a framework named Qwen1.5-0.5B for automatic extraction of 14 chest X-ray pathologies from radiology reports by tuning the framework with instructions. The combination of MedCLIP retrieval mechanisms further decreased the error rates in noisy clinical reports, following fine-tuning with supervised instruction learning on MIMIC-CXR. However, computational cost and hallucination are still significant problems. The 21st Century Cures Act has placed greater emphasis on improving the understanding of patients' radiology reports. Amin et al. [19] studied the use of AI to simplify with a combination of ChatGPT and ImpressionGPT for creating patient-friendly reports. They were found to be more easily readable by evaluation using Flesch-Kincaid and Gunning Fog metrics, but had errors between 3-6% on clinician evaluation, highlighting the need for clinician supervision. Likewise, Brian et al. [20] conducted a survey of automated biomedical text simplification methods, and classified them into rule based, neural and hybrid methods. The study revealed the weakness of the standard metrics like BLEU and SARI to assess clinical semantic preservation and pointed out that the support of multilingualism and low-resource learning are relevant research directions for the future.

B. Evolution of Deep Learning-Based Report Generation

The generation of the radiology report from images has been traditionally stated as an image to sequence problem. Li et al. [2] introduced a generic architecture consisting of image encoders, enhancement modules and language decoders. Their review included the shift from CNN-LSTM models to pretraining models like CLIP and BLIP-2, which are based on Transformer architectures.

Pang et al. [22] also reviewed the development of deep learning techniques in medical report generation. The survey explored the following methods: hierarchical recurrent networks, attention mechanisms, and reinforcement learning. These methods were studied using benchmark datasets such as IU X-Ray, MIMIC-CXR, and ChestXray14. The authors found that these linguistic measures, including BLEU, METEOR and CIDEr, lack the precision to determine diagnostic accuracy, and called for increased collaboration between AI researchers and clinicians.

Meléndez et al [21] recently conducted a scoping review of 89 Automatic Radiology Report Generation (ARG) studies, published from 2015 to 2025. The study found that CNN-Transformer networks are the most common architecture, and found significant modality and geographical biases in current datasets. The study suggested the use of more clinically relevant metrics like RadGraph and RadCliQ instead of traditional linguistic ones to better support evaluation in the real world.

C. Impact of Large Language Models in Radiology

The use of Large Language Models in radiology workflow is growing. Nakaura et al. [5] assessed the applications of GPT-4, BERT, and Transformer models in report generation, protocol

TABLE I
AI-BASED RADIOLOGY REPORT GENERATION AND MULTIMODAL MODELS

Ref	Method / Model	Dataset	Key Metrics	Year	Limitations
[18]	Qwen1.5-0.5B + SFT + MedCLIP RAG	MIMIC-CXR	Weighted F1 = 0.9014	2025	Hallucination risk; high computational cost
[19]	ChatGPT, ImpressionGPT for report simplification	Radiology reports	Flesch-Kincaid; 3–6% error rate	2023	Requires clinician supervision
[20]	Rule-based, Neural and Hybrid BTS methods	45 reviewed studies	BLEU, SARI, FKGL	2022	Metrics fail to capture clinical accuracy
[1]	Vision-Language Models, CLIP-style learning, RAG	18 datasets including MIMIC-CXR and VQA-RAD	BLEU, ROUGE	2024	Lack of universal multimodal metrics
[2]	CNN/ViT encoders with Transformer decoders	MIMIC-CXR, IU X-Ray	BLEU, ROUGE	2025	Hallucinated findings; dataset bias
[21]	ARRG scoping review	89 studies (2015–2025)	RadGraph, RadCliQ	2026	Geographic and modality bias
[5]	GPT-4, BERT and multimodal LLMs	Multi-domain clinical data	Board examination benchmarks	2024	Hallucination and explainability issues
[22]	CNN-RNN, Attention and RL-based report generation	IU X-Ray, MIMIC-CXR, ChestXray14	BLEU, METEOR, CIDEr	2023	Linguistic metrics do not reflect diagnosis
[3]	ViGPT2 + Cross-Attention + RAG	Indiana University Open-I	BLEU-1 = 0.571; Greedy Matching = 0.9882	2024	Single dataset validation
[23]	Learned Knowledge Base with Multi-modal Alignment	MIMIC-CXR, IU-Xray	BLEU, CIDEr, CE scores	2023	Difficulty detecting subtle abnormalities
[4]	MAIRA-1, ClinicalBLIP, LLaVA-Med	MIMIC-CXR, VQA-RAD	RadGraph-F1, RadCliQ	2025	Dataset scarcity; catastrophic forgetting

selection, longitudinal summarization, and board examination preparation. The study highlighted that while LLMs have demonstrated remarkable abilities, challenges such as hallucination, explainability, and data leakage limit its application in clinical settings. Good supervision is therefore essential at all times.

D. Multimodal Vision-Language Models

The integration of images and clinical text has been made possible by the advent of Vision-Language Models (VLMs). Hartsock et al. [1] compiled a list of medical VLM architectures and divided them into early-fusion and late-fusion categories. Three key areas in the survey that stood out as key drivers of performance improvement were contrastive learning, self-supervised learning, and Retrieval-Augmented Generation (RAG). Raminedi et al. [3] presented a multimodal architecture that uses Vision Transformers and GPT-2 with cross-attention mechanisms. The use of Chroma Vector Store and LangChain-based RAG allowed for the retrieval of clinically similar cases and minimizing hallucination. The framework was evaluated on the Indiana University Open-I dataset, with results of BLEU score 0.571 and Greedy Matching 0.9882, which shows high level of semantic consistency. Shuxin et al. [23] proposed a learned knowledge-base framework that aligned the visual features with textual embeddings and disease labels triple-modally. Its accuracy in linguistic and clinical performance scores surpassed previous papers like R2Gen and PPKED when tested on MIMIC-CXR and IU-Xray datasets.

E. Multimodal Large Language Models in Radiology

Multimodal Large Language Models (MLLMs) are the next generation of AI for radiology. Yi et al. [4] provided a survey on MAIRA-1, ClinicalBLIP, and LLaVA-Med for advanced systems in Radiology Report Generation and Radiology Visual Question Answering. The authors showed that radiology-specific metrics (RadGraph-F1, RadCliQ) are superior in their evaluation of clinical correctness compared to other linguistic measures. Several future advances are anticipated including the Retrieval-Augmented Generation, Reinforcement Learning from Human Feedback, and federated learning techniques.

IV. CLINICAL NLP AND PATIENT-CENTRIC APPLICATIONS

A. Ontology-Driven Rare Disease Identification

Structured coding systems often do not adequately represent rare diseases, which can make the identification of these diseases from clinical narratives difficult. To overcome this difficulty, Dong et al. [12] introduced a two-stage approach for rare disease recognition based on unstructured clinical notes and ontology. The framework was applied to the MIMIC-III and NHS Tayside datasets and resulted in precision gains of 30–50% compared to baseline methods. Weak supervision was shown to be a powerful approach for discovering hidden disease cases that are not covered by the traditional ICD coding system.

B. Cross-Institutional Generalization of Transformer Models

Transformative clinical NLP models may perform very well in a single institution but have showed to be transferable

TABLE II
CLINICAL NLP AND PATIENT-CENTRIC APPLICATIONS

Ref	Method / Model	Dataset	Key Metrics	Year	Limitations
[12]	UMLS + ORDO ontology with BlueBERT weak supervision	MIMIC-III, NHS Tay-side	Precision improvement = 30–50%	2023	Limited ontology coverage; abbreviation ambiguity
[24]	Fine-tuned BERT and SentenceBERT	EHR clinical notes	Cross-institution performance analysis	2023	Writing-style variation affects generalization
[13]	GPT-4o for Patient-Friendly Clinical Notes	Emergency Department notes	Understandability = 87.2%; PPQ 40→56.6	2026	Hallucination risk; clinician validation required
[14]	PubMedBERT, ClinicalBERT and RoBERTa for information extraction	34 studies (2018–2023)	Encoder models outperform GPT-4	2024	Lack of external validation
[15]	GPT-4, Claude and Llama with RAG	UpToDate benchmarks	Accuracy improvement = 16.6%	2025	HIPAA and accountability concerns

across other healthcare environments are still an issue. Kevin et al. [24] explored the robustness of fine-tuned BERT models for extracting outcomes related to epilepsy from EHRs. They found that the models could well be generalised within the same specialty but that performance drops rather drastically between providers and specialties. Training through multiple institutions and domain adaptation were highlighted as critical approaches to enhance scalability.

C. Patient-Friendly Clinical Note Generation

Increasing patient engagement is a key goal of health communication. Kim et al. [13] investigated how to use GPT-4o to produce Patient-Friendly Clinical Notes (PFCNs) for Emergency Department contexts. The notes created had a score of 87.2%, and the Patient Participation Questionnaire scores rose from 40 to 56.6. The outcomes presented here suggest that leveraging LLM summarization can greatly improve patient engagement and communication.

D. Information Extraction from Radiology Reports

The Radiology Report is a crucial component that provides valuable insights for clinical research, quality assurance, and AI model development. Reichenpfader et al. [14] conducted a scoping review of 34 papers that studied information extraction from radiology reports using LLM methods. The review confirmed that early generative models could not be compared to the encoder-based models (e.g., PubMedBERT, ClinicalBERT, and RoBERTa) in the context of structured extraction. The current literature primarily focused on thoracic and neurological imaging. Some of the major drawbacks were the lack of external validation, varying methods of annotation, and poor generalization across institutions.

E. Large Language Models in Medical Education and Healthcare Administration

In addition to diagnostic use cases, LLM are being used in healthcare education, clinical decision support, and healthcare administration. Vrdoljak et al. [15] conducted a thorough analysis of GPT-4, Claude and Llama-based systems in these areas. The study also emphasized on an important mechanism

to enhance reliability, which is Retrieval-Augmented Generation: The model with grounding on authoritative medical resources outperformed the other model by 16.6% accurate.

V. HEALTHCARE INTEROPERABILITY AND SEMANTIC STANDARDIZATION

A. FHIR-Based Clinical Data Standardization

The papers [6]–[9] are concentrated on healthcare interoperability with HL7 FHIR and other standards. To address this, Alberto et al. [6] suggested a modular ETL system that covered 98.7% of the fifteen FHIR resource types. Bossenko et al. [7] presented a low code CDA-to-FHIR transformation framework that relies on reusable visual components. The interoperability was developed by Marino et al. [8] in a systematic taxonomy and the importance of technical, organizational and territorial integration was pointed out. Raoof et al. [9] listed seventy-three problems that they saw in the adoption of FHIR, and highlighted SMART-on-FHIR and cloud platforms as key enablers. Collectively, these studies establish FHIR as the foundation for AI-ready healthcare infrastructures.

B. Ontology Integration and Semantic Knowledge Representation

The meaning of information exchange must be preserved, even if they are semantic consistency. However, a contextual modeling approach to mapping integrated RadLex and SNOMED CT, with a success rate of 99.6% for procedures related to x-rays [10]. A semantic web framework has been proposed by Zhang et al. [11] to integrate the SemMedDB database, KEGG and PharmGKB by using RDF triples and UMLS identifiers, which resulted in the identification of twenty-five novel disorder-gene-drug relationships with an accuracy of 89.88% validated by experts. The studies show how essential ontology-based systems are to the support of precision medicine and biomedical knowledge discovery.

C. Privacy-Preserving Healthcare Data Sharing

Ensuring patient privacy is paramount in cross-institutional AI applications. S. Phuyal et al. [26] proposed an architecture that uses blockchain technology, Verifiable Credentials and Zero-Knowledge Proofs to provide secure exchange of health

TABLE III
HEALTHCARE INTEROPERABILITY AND SEMANTIC STANDARDIZATION

Ref	Method / Framework	Dataset / Validation	Key Metrics	Year	Limitations
[6]	FHIR-based ETL pipeline	MOTU dataset	98.7% resource coverage	2025	Manual mapping expertise required
[7]	CDA→FHIR transformation using FML and TermX	Estonian NHIS	Reduced technical debt	2024	Complex mappings require domain expertise
[8]	Four-facet interoperability taxonomy	370 studies (PRISMA)	Systematic taxonomy analysis	2025	Integration complexity across systems
[10]	RadLex + SNOMED CT integration	2341 procedures	99.6% X-ray mapping; 47% angiography mapping	2025	Low performance for complex modalities
[26]	Blockchain + ZKP + FHIRopenEHR framework	Synthetic ICU dataset	98.5% mapping success	2026	Scalability challenges
[9]	FHIR adoption scoping review	Global literature	73 challenges identified	2024	Legacy system integration barriers
[27]	FHIR-DIET de-identification framework	FHIR resources	~400 requests/sec	2024	Re-identification risks
[25]	FHIR + DICOMweb + SVG imaging workflow	Imaging workflow validation	Browser-native interoperability	2023	CAD-to-FHIR integration complexity
[11]	RDF Semantic Web framework	SemMedDB, KEGG, PharmGKB	89.88% expert accuracy	2020	Dependent on source data quality

TABLE IV
AI ETHICS, TRANSPARENCY AND REPORTING STANDARDS

Ref	Framework / Method	Dataset / Scope	Key Metrics	Year	Limitations
[16]	GPT-4 Turbo report simplification framework	500 radiology reports	~20% errors below Grade 11 readability	2025	Risk of oversimplification and information loss
[17]	STARD-AI reporting guideline	240+ stakeholders	40 reporting items (18 new/revised)	2025	Requires widespread adoption

data with a mapping success rate of 98.5% between standards. E. Raso et al. designed an FHIR-DIET de-identification framework that is consistent with GDPR and HIPAA and can process up to 400 requests per second [27]. These frameworks give robust privacy assurances and facilitate secondary use of health data.

D. Modern Imaging Workflows and AI Integration

In order to overcome the problems of the legacy workflow, Tang et al. [25] introduced a FHIR- and DICOMweb-based imaging architecture. The framework allows for AI-generated findings to be used as interoperable clinical resources, using SVG-based annotations and conversion of CAD outputs to FHIR DiagnosticReports. These architectures will form the basis for future Precision Medicine and Intelligent Imaging Ecosystems.

VI. AI ETHICS, TRANSPARENCY AND REPORTING STANDARDS

With the increased adoption of AI systems in clinical practice, it is crucial to promote transparency, fairness, and patients' safety. Lee et al. [16] explored the ethics of balancing readability and clinical accuracy in AI-driven radiology report simplifications. They found that when they were very simplified, it could be clinically significant, so some oversight from clinicians and dual reporting strategies are needed. In order to increase the reproducibility and accountability, Sounderajah

et al. [17] proposed the STARD-AI framework for reporting AI-based diagnostic studies, which consists of 40 items. These studies collectively highlight the need for transparency, validation of fairness in AI deployment, and standardized reporting in responsible AI practices.

VII. CROSS-CUTTING CHALLENGES

Even with improvements made, there are still some challenges that are still impeding the clinical use of AI systems. Hallucination is still a significant issue in the generation of reports and summarizing patients. Current evaluation tools like BLEU and ROUGE can not fully capture the diagnostic accuracy. The model generalization is further impacted by cross-institutional degradation due to reporting style and clinical practices. Moreover, inconsistencies between standards like FHIR and SNOMED CT, and RadLex further hinder interoperability. Preserving privacy, complying with regulations, and having small, multilingual datasets are also significant challenges to widespread adoption.

VIII. FUTURE RESEARCH DIRECTIONS

Future studies should be directed towards the development of clinically useful evaluation indices that better represent diagnostic accuracy. Structured ontologies can help to enhance factual consistency and minimize hallucinations when combined with Large Language Models. The Federated

Learning frameworks can improve privacy and support multi-institutional cooperation. The automated ontology mapping and intelligent FHIR transformation tools can help to minimize manual domain expertise. Moreover, the ability to deploy AI systems across multilingual and low-resource healthcare settings will be crucial in the quest for equitable and globally accessible clinical AI

IX. CONCLUSION

This survey took a approach to the recent advancements of multimodal AI in radiology and health informatics, combining 27 peer-reviewed studies that were synthesized. The literature reviewed shows significant advancements in automated report generation, clinical NLP, interoperability frameworks, and ethical AI governance. Despite their good performance, hallucination, semantic consistency, privacy, and clinical reliability issues still obstruct the wide-spread use of modern Transformer-based architectures and multimodal models. Standardized evaluation, ontology-based learning, interoperable infrastructures, and reporting frameworks will be key to achieving clinically safe and trustworthy AI systems in the future healthcare ecosystem.

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